

Microsoft Access Homework: On-Campus Health Database

You will be assisting "On-Campus Health," the student infirmary at the University of Denver. Currently, their meticulous pharmacy recordkeeping is inefficient and costly due to manual systems and strict industry regulations like HIPAA and state laws regarding prescription drugs. While they have started using an Excel spreadsheet, they need a robust system to maintain and store data properly.

The current workflow involves students presenting doctor's orders for prescriptions, and the pharmacist logging the drug, student name, and prescribing doctor into the system. You will use the provided data file, Hw_Data.xlsx, to complete this system transition.

Answer Sheet Submission Instruction

Students must download the answer sheet from under Homework 2, complete it, and upload the filled file back to this Workshop. The uploaded file name must follow this exact format:

GroupNo#_Answersheet.docx

For example: GroupNo3_Answersheet.docx

Please make sure to use your correct group number in the file name before uploading.

Teamwork & Registration Requirement

Group Size: You will complete this assignment in teams of **4 students**.

Registration Deadline: Each team must register in the "**Groups**" section of the course system no later than **Friday at 22:00**.

Random Assignment: Students who do not register for a group by the deadline will be **randomly assigned** to a team by the instructors.

Technical Recommendation: Since this homework requires the use of **Microsoft Access** and must be completed on a **Microsoft/Windows computer**, it is strongly recommended that **at least one member of each group have access to a Microsoft computer**. Alternatively, students may use the **university lab computers**, which provide the necessary software environment.

Critical Assignment Rules

- **Strict Naming Convention:** You must not change any table names, column IDs, or attribute names provided in the Excel file. Any changes to these exact names will result in a grade of zero.
- **Access Implementation is Mandatory:** For all steps, the work must be properly created inside your Microsoft Access files. If a required element is not built in the database, you will receive zero points for that section, even if you wrote the correct answer in your Word document.

Part 1: Relational Database Development

Step 1: Database Setup (10 points)

Create an empty Microsoft Access relational database named hw2_group_NO#_relational.mdb. Import the provided Excel data, assigning the correct fields, data types, and primary keys for each table. Record your answers regarding this step in Part (a) of the provided Word document.

Step 2: Relationships (10 points)

Establish the necessary relationships between your tables in the Access database. Document these connections in Word document Part (b), clearly indicating the specific type of each relationship.

Step 3: SQL Queries (15 points total; 5 points per query)

Use the Query Design tool in Access to build the following three queries, then copy the resulting SQL View text into the corresponding sections of your Word document.

- Query 1 (Word Part g): Create a list showing student genders along with the total cost and average cost of their prescribed medications. Group the results by gender and sort them by the total cost in descending order. Switch to SQL View and copy the text.
- Query 2 (Word Part h): Create a list of student names alongside the total cost and count of medications they have been prescribed. Group the results by student name and sort the list in descending order by total cost. Switch to SQL View and copy the text.
- Query 3 (Word Part i): Create a list displaying the Doctor ID, Drug UPN, Prescription ID, Student ID, Date, Cost, Price, and Quantity. Add a calculated field named "Profit" that subtracts the cost from the price, and then multiplies that result by the quantity. Switch to SQL View and copy the text.

Part 2: Dimensional Database Transformation (Data Warehouse)

Transform your existing Microsoft Access relational database into a dimensional database optimized to analyze Profit and Loss for medications. You must appropriately transform, clean, and organize the data to allow for multi-dimensional analysis.

Step 4: Dimension and Fact Mapping (20 points total)

Your new structure must adhere to dimensional design principles, such as a Star or Snowflake Schema, to support business intelligence.

- Facts (10 points): In your Word document Part (c), list the individual facts.
- Dimensions (10 points): Underneath the facts in Word document Part (d), list the supporting dimension tables. Ensure your design accounts for Students (Demographics/Patients), Doctors (Prescribing Physicians), Drugs (Product details), Receipts/RX (Transaction specifics), and Time Period (Hierarchies for Month, Quarter, and Year).

Step 5: Dimensional Table Setup (10 points)

Create the new dimensional database named hw2_group_NO#_dimensional.mdb. Build the tables in Microsoft Access with the appropriate fields, data types, and primary keys. Record these structural details in Word document Part (e).

Step 6: Dimensional Relationships (10 points)

Create the necessary relationships for your new dimensional database inside Access. Document these relationships and clearly specify their types in Word document Part (f).

Step 7: Dimensional Database Query (5 points)

Use the Query Design tool in your new dimensional database to build the following analytical query, then copy the resulting SQL View text into your Word document.

- Query 4 (Word Part j): Create a query that calculates the total profits, total cost, and the count of drugs given to students, broken down by quarter for the year 2007. Once the query is built, switch to SQL View and copy the resulting SQL text into Part (j).

Submission Requirement (IMPORTANT)

You must upload a total of 3 files to this Workshop. Submissions that do not include all required files will be considered incomplete.

Please make sure your files follow the exact naming format below:

hw2_group_No#_relational.mdb

hw2_group_No#_dimensional.mdb

GroupNo#_Answersheet.docx

Replace No# with your actual group number (e.g., No3).